Experiment-4(8 queen problem solving)

Aim:
To place 8 queens on a chessboard such that no two queens attack each other, using backtracking.

Algorithm:

Represent the chessboard as a list where each index corresponds to a row and the value at that index is the column position of the queen.

For each row, try placing a queen in every column.

Before placing, check if the position is safe:

No queen in the same column.

No queen in the same diagonal.

If safe, place the queen and recursively solve for the next row.

If all rows are filled, a solution is found.

If no valid position exists, backtrack and try another column.

Python Program:

python

```
N = 8
board = [-1] * N

def is_safe(row, col):
    for i in range(row):
        if board[i] == col:
            return False
        if abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve(row):
    if row == N:
        return True
    for col in range(N):
        if is_safe(row, col):
            board[row] = col
            if solve(row + 1):
```

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```
            return True
        board[row] = -1
    return False

if solve(0):
    for row in range(N):
        for col in range(N):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print("-", end=" ")
        print()
else:
    print("Solution not found")
```

Sample Output (One Possible Solution):

Code:

```
Q - - - - - - -
- - - - Q - - -
- - - - - - - Q
- - - - - Q - -
- - Q - - - - -
- - - - - - Q -
- Q - - - - - -
- - - Q - - - -
```